

Sexism Detection with Learning from Disagreements

Ali Akbarhalvaei (0001113569)

Saleh Mir Mohammad Rezaei (0001157141)

Master's Degree in Artificial Intelligence, University of Bologna
{ali.akbarhalvaei, sa.mirmohammadrezaei}@studio.unibo.it

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Project Title

Learning from Disagreement: A Multilingual Ensemble Approach for Sexism Detection in Social Media

Abstract

The detection of sexist content in online platforms is a complex and inherently subjective task, as different individuals may interpret the same content differently. Traditional machine learning approaches typically rely on aggregated labels, thus ignoring the variability and disagreement among annotators.

This project aims to address sexism detection under the *learning from disagreement* paradigm, where models are trained directly on annotator-level annotations represented as probability distributions (soft labels), rather than single ground-truth labels.

The project proposes a multilingual system based on transformer models (e.g., mBERT and XLM-RoBERTa) combined through an ensemble strategy. The system will address multiple levels of analysis, including: (i) binary sexism detection, (ii) intention classification, and (iii) fine-grained sexism categorization.

The project will focus on predicting probability distributions that reflect annotator disagreement, comparing them with traditional hard-label approaches, and evaluating their effectiveness in capturing the subjective nature of sexism in social media content.

Description: Goals and Expected Deliverables

The project includes study, design, and implementation phases. The goals are structured incrementally to ensure feasibility and adaptability in case of technical or computational constraints.

Goal 1: Dataset Understanding and Preprocessing

Study and preprocess the EXIST dataset, which includes multilingual (English and Spanish) tweets annotated by multiple annotators.

- Extract and represent annotator labels as probability distributions (soft labels)
- Output: cleaned and structured dataset, preprocessing scripts

Goal 2: Baseline Model Implementation

Implement baseline transformer-based models (e.g., mBERT or XLM-RoBERTa) for sexism detection using standard (hard-label) training.

- Fine-tune models on training data
- Evaluate performance using classification metrics (e.g., F1-score)
- Output: baseline models and evaluation results

Goal 3: Learning from Disagreement (Soft Labels)

Extend the models to learn directly from soft labels.

- Train models to predict probability distributions over labels
- Use appropriate loss functions (e.g., cross-entropy with soft targets)
- Evaluate using soft metrics (e.g., cross-entropy, ICM-Soft)
- Output: trained soft-label models and comparative analysis

Goal 4: Ensemble Model Design

Develop an ensemble approach combining multiple models (e.g., mBERT and XLM-RoBERTa).

- Aggregate predictions via mean aggregation of probability distributions
- Compare ensemble performance with individual models
- Output: ensemble model and performance comparison

Goal 5: Evaluation and Analysis

Perform a comprehensive evaluation of the proposed system.

- Compare hard-label vs soft-label approaches
- Analyze the effect of modeling disagreement
- Evaluate performance across tasks and languages
- Output: experimental results, tables, and discussion

Development Methodology

The project will be developed using a structured workflow based on Git version control (e.g., GitLab flow). All code and experiments will be tracked in a dedicated repository, ensuring reproducibility and incremental development. Build automation and dependency management will be handled through standard Python tools.

Expected Deliverables

1. Software Artifacts:

- Python-based implementation using frameworks such as PyTorch and HuggingFace
- Clean and well-documented Git repository
- Reproducible training and evaluation pipeline with build automation

2. Final Report:

- Description of methodology and design choices
- Explanation of the learning from disagreement paradigm
- Experimental evaluation and results
- Discussion of limitations and possible improvements